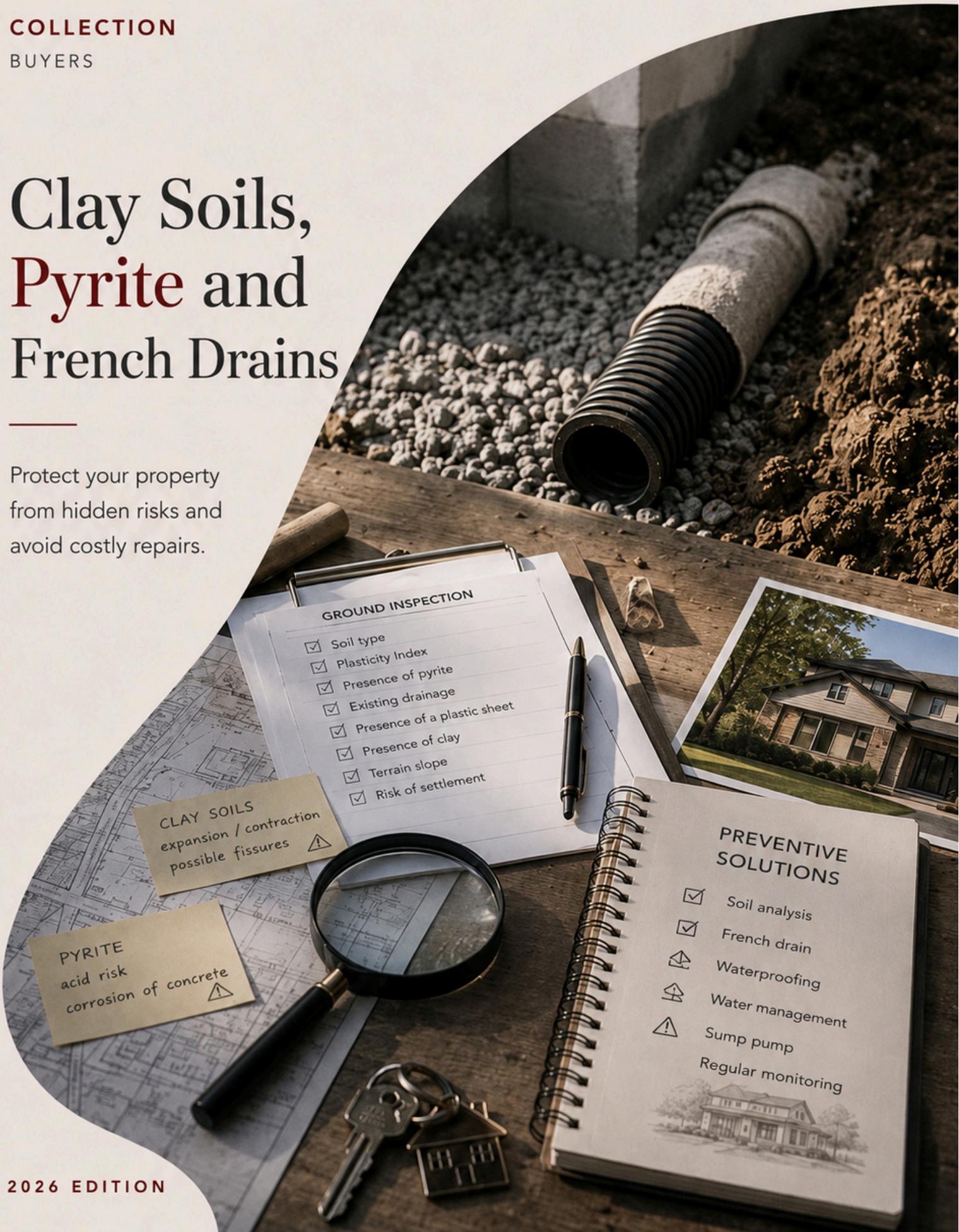
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Nº. 14

COLLECTION
BUYERS

Clay Soils, Pyrite and French Drains

Protect your property from hidden risks and avoid costly repairs.



2026 EDITION

BEFORE YOU BEGIN

Important notice

This guide is provided for informational and educational purposes only. It presents general information about real estate in Quebec and does not constitute personalized real estate, legal, tax or financial advice. Every real estate situation is different, and you should consult the appropriate professionals before making a decision about a transaction.

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COLLECTION BUYERS · GUIDE NO. 14

Clay soils, pyrite and French drains

The Québec-specific field checks — buying with your eyes open

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Certain building risks are specific to Quebec and the Montreal region: clay soils, pyrite, aging French drain, buried tanks, vermiculite. Well identified, they can be managed; ignored, they are expensive.

This guide teaches you how to spot the signals, ask the right questions and target professional checks before you buy.

This content is informative: diagnoses must be made by professionals (inspector, engineer, expert). Each chapter ends with a lined “My Notes” page.

Quebec buildings have their particularities: clay soils, pyrite and pyrrhotite, French drains, buried reservoirs. Well known, these issues can be managed; ignored, they are expensive. This guide teaches you how to spot them and protect yourself before you buy.

Objective: transform scary words into concrete verification points for your inspection.

CHAPTER

Clay soils and foundation movement

> *Understand why clay moves and what that implies.*

A large part of the region of Montreal, Laval and the North Shore is based on clay soils. Clay expands with water and shrinks in dry periods, which can cause foundation movement, cracks and uneven floors - especially during droughts or with large trees nearby drying out the soil.

- Signals: stepped cracks, sticking doors/windows, leaning floors.
- Aggravating factors: mature trees near foundations, poor drainage, droughts.
- If in doubt: have it assessed by a professional (sometimes an engineer).

Clay swells with water and contracts during dry periods, which can cause the foundations to move (cracks, leaning floors). This is not inevitable: good water management around the house (slopes, gutters, stable watering of foundations in drought) strongly limits movement.

> EMILIE'S ADVICE
>
> *Not all cracks are serious, but some are. In clay areas, have major cracks assessed by a professional before purchasing: it's the difference between a cosmetic detail and a major bill.*

CHAPTER

Pyrite and pyrrhotite

> *Recognize these mineral risks depending on the sector.*

Pyrite can be present in the backfill under the basement slabs in certain sectors: as it swells, it lifts and cracks the slab. Pyrrhotite concerns the foundation concrete of certain regions (especially outside Greater Montreal, but be aware) and can degrade it.

- Pyrite signals: lifting and cracking of the basement slab, pushing on the partitions.
- Verification: swelling test of the embankment (petrography) by a specialized laboratory.
- Certain sectors are more at risk: find out locally.

Pyrite (under slabs) and pyrrhotite (in concrete in certain regions) can, by reacting, crack and lift the concrete. The risk varies depending on the sector and the time of construction. Specialized tests exist: if in doubt, do them before buying.

> EMILIE'S ADVICE
>
> *If a basement slab shows cracking and heaving in an area known for pyrite, request testing before purchasing. The cost of the test is insignificant compared to that of a slab to be redone.*

CHAPTER

The French drain and water management

> *Evaluate the system that keeps the basement dry.*

The French drain, buried at the foot of the foundations, evacuates water from the ground. Over time (often several decades), it can become clogged or sag, causing infiltration into the basement. Its replacement is a costly project (excavation).

- Signals: humidity or infiltration in the basement, efflorescence, odors.
- Good surface drainage: slope of the land and gutters directing water away from the house.
- A camera can inspect the condition of the drain in case of doubt.

The French drain, buried at the foot of the foundations, evacuates water from the ground; blocked or collapsed, it allows water to infiltrate into the basement. Its replacement is expensive. Observe the clues (moisture, stains, odor) and ask for the age and history of the drain.

> EMILIE'S ADVICE
>
> *The age of the French drain is a question that should be asked systematically on a house over 30 years old. A dry basement today does not guarantee a drain in good condition.*

CHAPTER

Buried tanks, vermiculite and asbestos

> *Identify other Quebec vigilance points.*

- Buried oil tank (former oil heating): risk of soil contamination and insurability issue; its presence must be declared and managed.
- Vermiculite (granular insulation in some attics/old walls): may contain asbestos; to be tested before work.
- Other asbestos materials in older buildings: precautions during renovations.

None of these elements is necessarily prohibitive, but each must be identified, evaluated and, if necessary, costed before purchase.

Stay attentive to other points in Quebec: buried oil tank (contamination), vermiculite that may contain asbestos, and old materials at risk. These elements can be managed, but it is better to know them before buying than to discover them while renovating.

> EMILIE'S ADVICE
>
> *These points shouldn't make you shy away from all old houses - they should make you ask questions and focus the inspection. A well-understood house is a house that you buy with peace of mind.*

CHAPTER

Detect these risks and protect yourself

> *Check before buying.*

These floor and building problems are managed well... when they are detected in time. Your purchase protections:

- **Pre-purchase inspection** by a professional attentive to these regional issues.
- **Specialized tests** as needed (pyrite, pyrrhotite, soil).
- **Questions for the seller** and careful reading of his statement.
- **History** works and repairs.

An inspection condition in your promise to purchase gives you time to validate all of this.

Your best protection: an inspection by a professional attentive to these regional issues, targeted tests if necessary, and an inspection condition in your promise to purchase. A few hundred dollars of expertise can avoid tens of thousands in repairs.

> EMILIE'S ADVICE
>
> *Always add an inspection condition to your offer, especially in sectors known for these issues. A few hundred dollars of expertise can save you tens of thousands in repairs.*

IN PRACTICE

Action plan

> *Your next steps, concretely.*

This week

1. Identify if my targeted sectors are at risk (clay, pyrite).
2. List the signals to watch for during visits.

Within 30 days

1. Target relevant field verifications with the inspector.
2. Provide specialized tests (pyrite, drain, vermiculite) if necessary.

Then

1. Integrate any identified risks into my pricing and conditions strategy.

To protect myself

1. Add an inspection condition and target sector issues.
2. Provide specialized tests (pyrite, soil) in case of doubt.

REFERENCES

Resources & glossary

- Clay soil: soil which swells and contracts, which can move the foundations.
- Pyrite: mineral from the backfill which swells and raises the basement slab.
- Pyrrhotite: mineral that can degrade foundation concrete (certain regions).
- French drain: pipe at the foot of the foundations evacuating water from the ground.
- Vermiculite: old granular insulation that may contain asbestos.
- Buried tank: old fuel oil tank, soil and insurance issues.
- Clay soil: soil that swells and contracts depending on the humidity, which can move the foundations.
- Pyrite / pyrrhotite: minerals that can cause concrete to swell or crack.
- French drain: peripheral drain evacuating water around the foundations.
- Buried tank: old fuel oil tank, risk of contamination.
- Vermiculite: insulation that may contain asbestos.

GO FURTHER

Quiz - test yourself

1. Clay soils can cause:

A) Nothing

B) Foundation movements and cracks

C) Better insulation

D) Energy savings

2. An aggravating factor in clay soil is:

A) A small lawn

B) Large trees near the foundations

C) A new roof

D) A new door

3. Pyrite mainly affects:

A) The roof

B) The backfill under the basement slab

C) Windows

D) Electricity

4. To check the pyrite, we do:

A) A photo

B) A swelling test (petrography) in the laboratory

C) Nothing

D) A water test

5. The French drain is used to:

A) Heating the house

B) Evacuate water from the ground at the foot of the foundations

C) Insulate the roof

D) Supply water

6. The age of the drain is a key question on a house over:

A) 5 years

B) 10 years

C) 30 years

D) only 100 years

7. Vermiculite may contain:

A) Gold

B) Asbestos

C) Drinking water

D) Concrete

8. A buried oil tank is above all an issue of:

A) Decoration

B) Soil contamination and insurability

C) Free heating

D) Nothing

Answers

1. B - Clay expands and contracts (chap. 1).

2. B - Trees dry out the ground (chap. 1).

3. B - It inflates and lifts the slab (chap. 2).

4. B - Specialized analysis of the embankment (chap. 2).

5. B - It keeps the basement dry (chap. 3).

6. C - It ages over a few decades (chap. 3).

7. B - To be tested before work (chap. 4).

8. B - Environmental and insurance risk (chap. 4).

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You now have the right keys in hand.

Every real estate journey is different.

If you would like to go further,
I would be happy to discuss
your project with you.



Make informed decisions.
Choose with confidence.
Live without regrets.
See you soon,
Emilie ☺

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